



GENERATIVE BIONICS

Analisi Strategica del Fit con i Bandi Horizon Europe
2026-2027

Cluster 2 – Culture, Creativity and Inclusive Society

Generato il 15/04/2026 01:14
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Executive Summary

Fit medio cluster

48.27%

Call analizzate

15

Top 3 call

#1 · HORIZON-CL5-2026-10-D6-03

GO

Generative AI for smarter CCAM: enhancing perception, decision-making, and validation (CCAM Partnership)

Score: 60.46%

#2 · HORIZON-CL5-2026-10-D6-02

GO

Geopolitical competition and socioeconomic resilience in CCAM: an innovation and policy roadmap for EU leadership (CCAM Partnership)

Score: 57.84%

#3 · HORIZON-CL5-2026-09-D4-01

WATCH

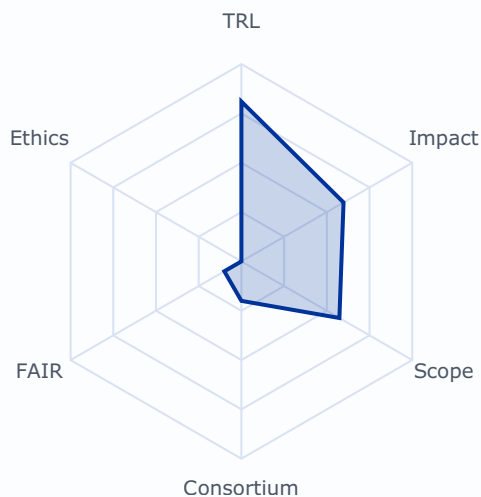
Researching the technical, social & economic factors impacting the energy performance of Smart Buildings (Built4People Partnership)

Score: 54.84%

Distribuzione Top 10

#1		60.5
#2		57.8
#3		54.8
#4		53.8
#5		53.7
#6		49.9
#7		47.7
#8		46.5
#9		45.2
#10		44.7

Radar complessivo cluster



Rank #1 · HORIZON-CL5-2026-10-D6-03

GO

Generative AI for smarter CCAM: enhancing perception, decision-making, and validation (CCAM Partnership)

Fit: 60.46 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 60/100

La call HORIZON-CL5-2026-10-D6-03 ottiene un fit del 60.5% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 30.4/100). Solver: Optimal, CR AHP: 0.0168.

"Project results are expected to contribute to all the following expected"

"Availability and integration of advanced, trustworthy, energy-efficient perception"

"systems, exploiting technological advancements of Generative AI (GenAI) to enhance"

Gap principali da colmare

- ⚠ Nessun gap bloccante evidente dai vincoli hard del solver

Azioni concrete (prossimi 3-6 mesi)

- ✅ Consolidare la bozza di proposal con piano operativo e risk register

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

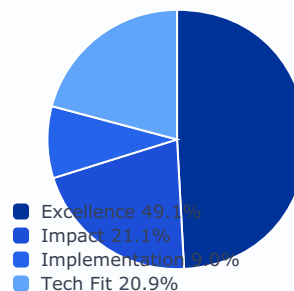
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

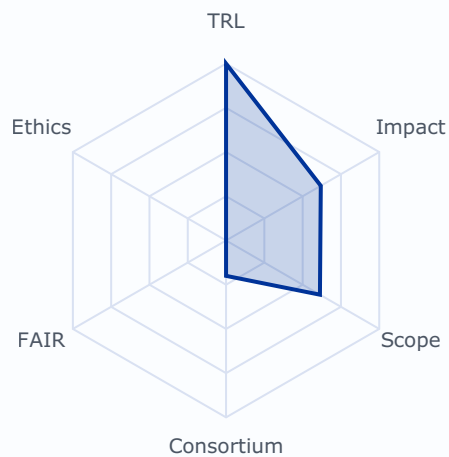
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.6185	0.6185	0.3039
impact	0.2105	0.6511	0.6511	0.1370
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.6129	0.6129	0.1278

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** False
- **budget_company_available:** 0.0
- **budget_max:** 13000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** False
- **gender_balance_required:** False

Expected Outcomes

Project results are expected to contribute to all the following expected outcomes:

- Availability and integration of advanced, trustworthy, energy-efficient perception systems, exploiting technological advancements of Generative AI (GenAI) to enhance situational awareness and support safe decision-making;
- Enhanced Vulnerable Road User (VRU) safety, based on elevated, more temper-proof perception and understanding of their behaviour and intention predictions;
- Enhanced robustness of CCAM systems - both on-board and on the infrastructure side - in critical situations due to their training, virtual testing and validation in scenarios generated by GenAI, complementing existing scenario databases for the testing and validation of CCAM systems;
- Enhanced understanding of the relevance and limitations of using GenAI for CCAM;
- Tools and harmonised approaches for the use of GenAI in mobility technology development, training and validation, as well as for systemic applications such as traffic management and remote control, integrating them into existing approaches.

Scope

Pilots and demonstrations using Level 3 and 4 vehicle services face major challenges in perception and decision making, highlighting the necessity for low-latency solutions that enhance responsiveness and situational awareness in real-time operating conditions. This is especially relevant for driving in more complex environments like urban areas, where environmental variance is higher and where new scenarios can be regularly encountered. Furthermore, there is the need to limit the latency, bandwidth and energy use for on-board calculations, as well as the need to enhance the security, privacy and reliability (e.g. scene understanding and prediction of near-future scenario development). For rapid decision-making in interactions with VRUs, this is essential for implementing CCAM-enabled solutions and ensuring scalability.

283 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf

At the same time, developments of sector-agnostic technologies show advancements -such as GenAI- that can be beneficial for CCAM. First exploratory steps can be expected from a project funded under HORIZON-CL5-2023-D6-01-02284 regarding the potential in the virtual generation of edge cases, which could be used for the development, training, virtual testing and validation of CCAM systems.

Further advancements in GenAI applications specifically for the CCAM domain need to be developed, trained and validated²⁸⁵. Thus, proposed actions shall include approaches to exploit further technological advancements for CCAM. Major steps are needed to advance to highly advanced, ultra-safe, trustworthy and energy efficient real-time perception and decision-making systems for automated vehicles, specifically focusing on scalable solutions and the exploitation of GenAI. These advancements should leverage low latency systems or distributed computing resources to facilitate real-time processing, thereby improving system responsiveness and safety. This topic will thus contribute to the AI Continent Action Plan²⁸⁶ by fostering AI development and adoption in the

Rank #2 · HORIZON-CL5-2026-10-D6-02

GO

Geopolitical competition and socioeconomic resilience in CCAM: an innovation and policy roadmap for EU leadership (CCAM Partnership)

Fit: 57.84 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 58/100

La call HORIZON-CL5-2026-10-D6-02 ottiene un fit del 57.8% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 31.9/100). Solver: Optimal, CR AHP: 0.0168.

"Project results are expected to contribute to all the following expected"

"Assessment of Europe's CCAM position in global competition, identifying strategic"

"vulnerabilities, dependencies and opportunities across business models, supply chains,"

Gap principali da colmare

- ⚠ La call richiede copertura SSH esplicita

Azioni concrete (prossimi 3-6 mesi)

- ✅ Integrare partner SSH e work package dedicato a impatti sociali/adozione

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

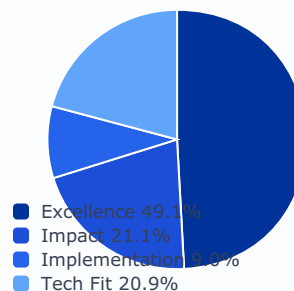
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

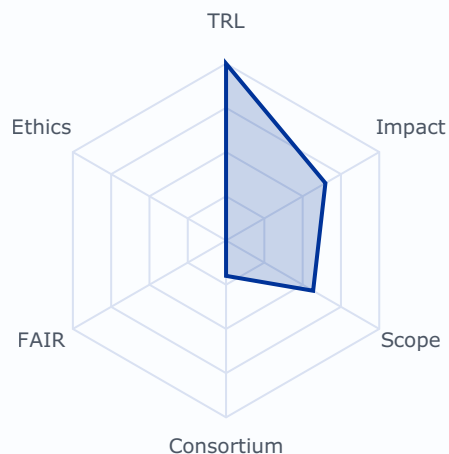
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.6487	0.6487	0.3187
impact	0.2105	0.6158	0.5000	0.1052
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.5685	0.5685	0.1186

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** False
- **budget_company_available:** 0.0
- **budget_max:** 4000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** True
- **gender_balance_required:** False

Expected Outcomes

Project results are expected to contribute to all the following expected outcomes:

- Assessment of Europe's CCAM position in global competition, identifying strategic vulnerabilities, dependencies and opportunities across business models, supply chains, critical components, technological capabilities, infrastructure, and regulatory frameworks, supported by advanced AI-driven analytics and innovation mapping tools.

- Stakeholder-driven, participatory future scenarios and strategic pathways that define Europe's leading role in the evolving geopolitical, technological, and economic landscape of CCAM. These should be developed using advanced foresight methods (e.g.

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qualitative scenarios, horizon scanning, technology roadmapping, etc.), complemented by iterative validation through a minimum of 3 dedicated Living Labs selected to represent diverse European regions, urbanisation levels, and governance capacities, ensuring a structured, robust, and anticipatory approach to long-term decision-making.

- A data-driven understanding of the socioeconomic effects of different CCAM deployment pathways, based on a quantified assessment across Member States and Associated Countries, economic sectors, and demographic groups, at national and cross-national levels, using innovative, integrated economic-transport modelling that captures dynamic interactions, systemic feedback loops, and long-term impacts. Socioeconomic effects may include, but are not limited to, employment and growth aspects, equity, and transport poverty.
- Robust governance models, policies and business strategies (including for SMEs and micro-enterprises) that strengthen Europe's leadership, economic resilience, and market positioning in CCAM, developed through interdisciplinary methods combining policy analysis, institutional diagnostics, and scenario-based

Scope

CCAM is a key area of global competition and one of the five pillars of the European Automotive Action Plan for the automotive sector, aimed at helping the industry regain its leadership in the shift towards smarter (AI-powered), cleaner, and more connected vehicles, However, CCAM deployment is shaped by evolving geopolitical dynamics, rapid technological advancements, and economic uncertainties. Europe must secure its leadership in CCAM and strengthen its socioeconomic resilience by continuously addressing vulnerabilities and identifying robust pathways for policy development and market deployment. This topic will assess the global geopolitical landscape of CCAM innovation, map future pathways, and develop evidence-based strategies for policymakers, businesses, and investors. The results will support robust, future-proof policies, business strategies, and investment frameworks, ensuring a resilient, inclusive, and competitive European CCAM ecosystem. Proposed actions are expected to address all of the following aspects:

- Conduct a comprehensive geopolitical and economic analysis of CCAM to map Europe's strategic position in global competition. Identify vulnerabilities, dependencies and opportunities in business models, supply chains, critical raw materials, technological capabilities, digital and physical infrastructure, validation processes, safety benchmarks,

and pricing strategies, supported by AI-driven analytics and innovation mapping tools.

The analysis should also assess large-scale CCAM initiatives worldwide, identifying scaling trajectories, tipping points, commercialization barriers, and success factors. These insights should inform strategic policymaking, investment decisions, and Europe's regulatory positioning in global CCAM markets.

- Develop novel, stakeholder-driven participatory future scenarios to explore plausible geopolitical, technological, and economic developments affecting CCAM, and define associated transition pathways toward desirable European futures that ensure resilience, strategic autonomy, and competitiveness. Use advanced foresight methods (e.g. qualitative scenarios, horizon scanning, technology roadmapping, etc.), complemented by iterative validation through a minimum of three

Rank #3 · HORIZON-CL5-2026-09-D4-01

WATCH

Researching the technical, social & economic factors impacting the energy performance of Smart Buildings (Built4People Partnership)

Fit: 54.84 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 55/100

La call HORIZON-CL5-2026-09-D4-01 ottiene un fit del 54.8% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 29.5/100). Solver: Optimal, CR AHP: 0.0168.

"Project results are expected to contribute to all of the following expected"

"221 This decision is available on the Funding and Tenders Portal, in the reference documents section for"

"Horizon Europe, under 'Simplified costs decisions' or through this link:"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance
- ⚠ La call richiede copertura SSH esplicita

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)
- ✅ Integrare partner SSH e work package dedicato a impatti sociali/adozione

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

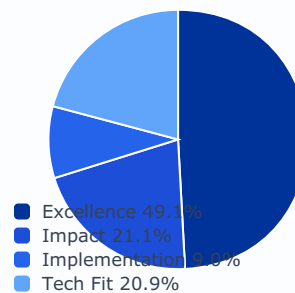
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

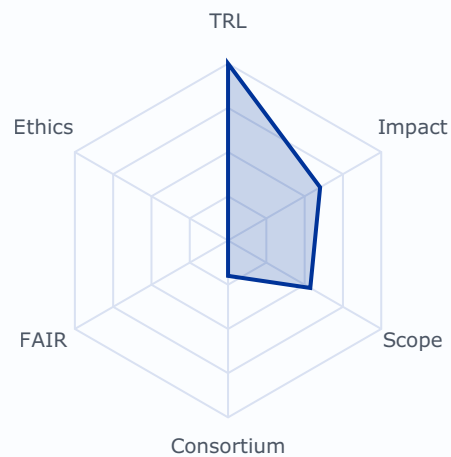
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.6011	0.6011	0.2954
impact	0.2105	0.5035	0.5000	0.1052
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.5368	0.5368	0.1120

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** False
- **budget_company_available:** 0.0
- **budget_max:** 15750000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** True
- **gender_balance_required:** False

Expected Outcomes

Project results are expected to contribute to all of the following expected outcomes:

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- Increased evidence of factors such as physiological, behavioural, social, environmental and cultural that influence how different user and demographic groups perceive and use smart, secured, integrated energy efficient building systems, and how it affects the building whole-life cycle and energy savings, as well as occupant satisfaction, health and well-being;
- The design and operation of smart building systems and smart buildings are improved making them more user-friendly and effective.

Scope

Significant investments have been made in developing hardware and software for smart buildings. There is still limited understanding how smart buildings solutions impact the energy performance of buildings and users' comfort in practice. Research is required on the technical, social, and economic factors that influence how different groups – defined by their social, educational, age, and financial status – use and interact with smart buildings and systems. The whole-life cycle impacts of smart building have to be better understood, from design and construction to operation, maintenance, end-of-life and circular re-use, to ensure delivery of sustainable and cost-effective smart building solutions.

Proposals are expected to address all of the following:

- Research and quantify how existing and novel smart building solutions meet the needs of different social and economic groups (based on their gender, age, educational and financial status, etc.);
- Identify major barriers and enablers for the use of smart building solutions among different socio-demographic and socio-economic groups, and research how it impacts the energy performance of buildings;
- Validate of findings in at least three different categories of existing buildings (e.g. private residential, social housing, commercial building, office building, etc.), each one in a different Member State or Associated Country.
- Develop actionable recommendations tailored to key stakeholders (such as equipment providers, architects, building owners and facility managers) to support the design, deployment and operation of inclusive, user-centric smart building solutions.

This topic requires the effective contribution of SSH disciplines and the involvement of SSH experts, institutions as well as the inclusion of relevant SSH expertise, in order to produce meaningful and significant effects enhancing the societal impact of the related research

Rank #4 · HORIZON-CL5-2026-03-D3-21

WATCH

Novel solutions for off-grid storage of renewable energy for critical infrastructures

Fit: 53.81 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 54/100

La call HORIZON-CL5-2026-03-D3-21 ottiene un fit del 53.8% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 29.2/100). Solver: Optimal, CR AHP: 0.0168.

"Project results are expected to contribute to all the following expected"

"Enhanced European knowledge and leadership in off-grid clean energy storage"

"Improved energy security and cost-effectiveness of Europe's energy systems, particularly"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

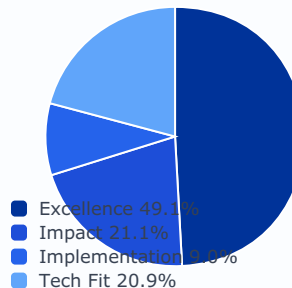
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

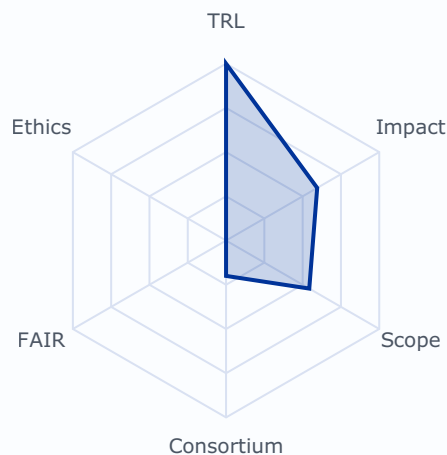
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.5950	0.5950	0.2924
impact	0.2105	0.4589	0.4589	0.0966
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.5432	0.5432	0.1133

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** False
- **budget_company_available:** 0.0
- **budget_max:** 12000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** False
- **gender_balance_required:** False

Expected Outcomes

Project results are expected to contribute to all the following expected outcomes:

- Enhanced European knowledge and leadership in off-grid clean energy storage technologies.
- Improved energy security and cost-effectiveness of Europe's energy systems, particularly for critical infrastructure.
- Elimination of fossil-based backup solutions for critical infrastructures, strengthening European competitiveness.

Scope

The topic aims to develop novel, cost-efficient off-grid energy storage solutions for renewable energy, tailored for critical infrastructures such as hospitals, transport hubs, data centres, and utilities. Solutions should address challenges specific to off-grid or non-interconnected area contexts, including storage scale, compromised grid interaction, resilience to potential shortages of critical materials, and multi-energy needs (electricity, heating, cooling). Emphasis should be on robustness and error-freeness, while considering sustainability, efficiency and performance. The project should foresee these solutions for using real-world critical infrastructures through simulation and experiments and establishing a procedure for decision-making, focusing on energy security and resilience against high-impact, low-probability events, by ensuring operational efficiency both on-grid and off-grid. Where relevant, interdependencies among critical infrastructures and system redundancy should be considered.

Projects are encouraged to support the reconstruction of Ukraine by including Ukrainian beneficiaries.

202 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf

Rank #5 · HORIZON-CL5-2026-10-D6-07

WATCH

Supporting sustainable and smart urban mobility in Europe (CIVITAS)

Fit: 53.68 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 54/100

La call HORIZON-CL5-2026-10-D6-07 ottiene un fit del 53.7% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 28.8/100). Solver: Optimal, CR AHP: 0.0168.

"Project results are expected to contribute to all of the following outcomes:"

"Valorisation, cross-fertilisation and dissemination of innovative solutions for sustainable"

"urban mobility, from EU and national projects through the CIVITAS initiative;"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance
- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)
- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

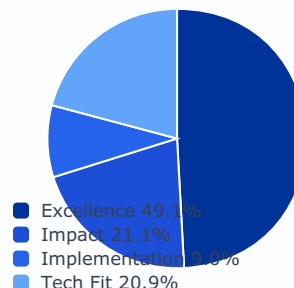
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

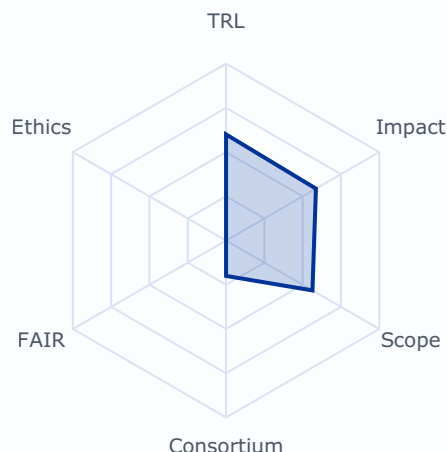
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.5868	0.5868	0.2883
impact	0.2105	0.4511	0.4511	0.0950
implementation	0.0896	0.4400	0.4000	0.0358
tech_fit	0.2086	0.5640	0.5640	0.1176

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call





Vincoli LP applicati e effetto

- **trl_violation:** False
- **budget_company_available:** 0.0
- **budget_max:** 4000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** False
- **gender_balance_required:** False

Expected Outcomes

Project results are expected to contribute to all of the following outcomes:

- Valorisation, cross-fertilisation and dissemination of innovative solutions for sustainable urban mobility, from EU and national projects through the CIVITAS initiative;
- Contribution to the accelerated deployment of innovative urban mobility solutions, increasing competitiveness of European urban mobility actors while achieving European mobility policy objectives.
- Cities are equipped with a relevant knowledge, necessary soft and technical skills to successfully deploy smart and sustainable mobility solutions, for passenger and freight.
- A vibrant community of urban innovative mobility stakeholders is maintained and strengthened.

Scope

Effective and efficient urban mobility systems are the backbone of competitive cities, without them the economy comes to a standstill. While cities are going through a rapid digital, green and social transformation, local and regional authorities play bigger roles and take on

305 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

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greater responsibilities in contributing to the prosperity, climate-neutrality and competitiveness of the European Union. However, the deployment of innovative sustainable mobility solutions has been hampered by fragmentation of the support to cities at regional, national and EU level, lack of resources for experimentation with new approaches, insufficient knowledge on research results and innovative solutions, and difficulties for urban stakeholders to exchange knowledge, experiences and lessons learned, including providing feedback to policy makers and the research community.

The aim of this coordination and support action is to address the barriers to the deployment of innovative urban mobility projects' results through a comprehensive and collaborative approach, while further strengthening the European urban mobility innovation ecosystem.

Taking into account the previous CIVITAS coordination and support action 306 , in collaboration with, and avoiding overlaps with, the Cities Mission Platform³⁰⁷ and other relevant initiatives, the action is expected to address - in a clear and streamlined approach - all of the following aspects:

- Harvesting and spreading innovative urban mobility solutions, and speeding up the take-up of the related projects results by European cities by:
 - ☐ setting up capacity building and a replication programme, to facilitate twinning, mutual learning and peer-to-peer exchanges, which should include cooperation with industry, SMEs and start-ups, while ensuring a feedback to policy loop;
 - ☐ optimising the common, user-friendly CIVITAS project evaluation framework, with dedicated support for its implementation by the

Rank #6 · HORIZON-CL5-2026-03-D3-12

WATCH

Long-lifetime and optimised use of materials in recyclable Ag and In-free Si PV modules (EUPI-PV Partnership)

Fit: 49.90 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 50/100

La call HORIZON-CL5-2026-03-D3-12 ottiene un fit del 49.9% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 26.3/100). Solver: Optimal, CR AHP: 0.0168.

"Photovoltaic (PV) energy systems are one of the cheapest and fastest"

"growing sources of electricity generation, largely thanks to an important decrease in the cost"

"of solar modules in the last 10-15 years and to their simple installation. There is no PV"

Gap principali da colmare

- ⚠ Excellence sotto soglia competitiva: narrativa scientifica da rafforzare
- ⚠ Impact non ancora distintivo su outcomes e policy relevance

Azioni concrete (prossimi 3-6 mesi)

- ✅ Definire proof plan tecnico con milestone TRL e metriche di validazione (90 giorni)
- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

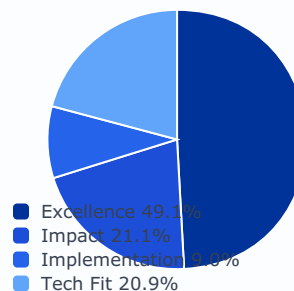
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

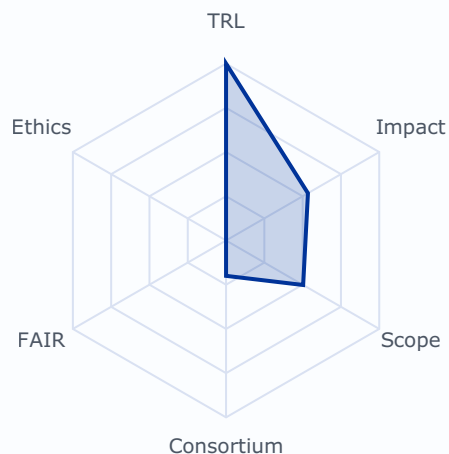
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.5347	0.5347	0.2627
impact	0.2105	0.4535	0.4535	0.0955
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.5033	0.5033	0.1050

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** False
- **budget_company_available:** 0.0
- **budget_max:** 15000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** False
- **gender_balance_required:** False

Expected Outcomes

Photovoltaic (PV) energy systems are one of the cheapest and fastest growing sources of electricity generation, largely thanks to an important decrease in the cost of solar modules in the last 10-15 years and to their simple installation. There is no PV lifetime definition, but manufacturers usually guarantee a 25-year lifetime with an expected degradation rate of 0.8% per year. Meanwhile, a surge in the number of end-of-life (EOL) PV modules is expected, of which crystalline silicon (c-Si) PV is the main type. It would therefore be beneficial to improve the eco-design and sustainability of silicon-based solar cells and modules, by reducing the use of Critical Raw Materials (CRMs), revalorising the use of recycled silicon waste, adopting recyclable/repairable module designs and extending their lifetime without compromising their reliability and durability.

Project results are expected to contribute to all of the following expected outcomes:

1. Ultra-low carbon footprint and reduced environmental impact PV cells/modules through lifetime extension, Ag or In free, and with designs that allow recyclability and repairability;
2. Support a European economic base which is stronger, more resilient, competitive, and fit for the green and digital transitions, by reducing strategic dependencies for critical raw materials and components;
3. Support the execution of the solar energy joint research and innovation agenda.

Scope

Proposals are expected to:

- Extend module lifetime up to 40 years with extreme weather resilience;
- Develop cell/module architectures/designs including recycled content, without In or Ag and without harmful substances (Pb, F, etc.) but without compromising their quality and reliability;
- Develop module designs with improved recyclability and repairability;
- Develop adapted testing methodologies and recycling standards;
- Provide technical solutions for PV module repair in the field to extend lifetime;
- Determine the environmental benefits of recycling precious metals.

This topic implements the co-programmed European Partnership for Innovation in Photovoltaics (EUPI-PV). As such, projects resulting from this topic will be expected to report on the results to the European Partnership for Innovation in Photovoltaics (EUPI-PV) in support of the monitoring of its KPIs.

Rank #7 · HORIZON-CL5-2027-02-D3-24

WATCH

Large scale operational validation and upscaling of state-of-the-art (Generative) AI tools and models powering a next generation digital energy system

Fit: 47.74 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 48/100

La call HORIZON-CL5-2027-02-D3-24 ottiene un fit del 47.7% dopo ottimizzazione LP. Il criterio dominante e' impact (contributo 16.9/100). Solver: Optimal, CR AHP: 0.0168.

"Project results are expected to contribute to all the following expected"

"Advanced, secure frugal (energy-, resource-, data- and cost-efficient, sustainable) AI"

"tools for real-time grid operations (including protection), market operations (if relevant)"

Gap principali da colmare

- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente
- ⚠ TRL aziendale inferiore al requisito della call (target: 7)

Azioni concrete (prossimi 3-6 mesi)

- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)
- ✅ Eseguire pilot dimostrativo per salire di almeno 1 livello TRL (3-6 mesi)

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

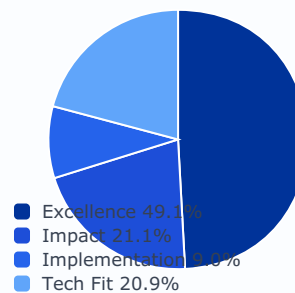
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

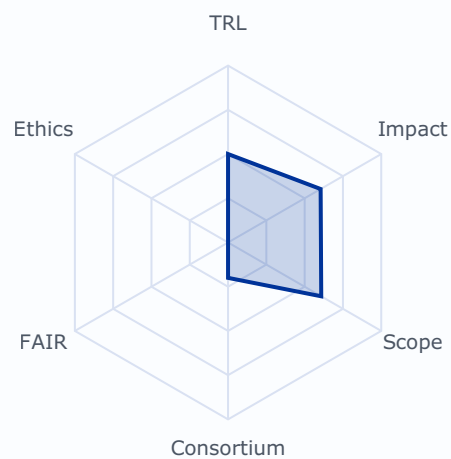
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.6044	0.3000	0.1474
impact	0.2105	0.8032	0.8032	0.1691
implementation	0.0896	0.3800	0.3800	0.0340
tech_fit	0.2086	0.6083	0.6083	0.1269

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** True
- **budget_company_available:** 0.0
- **budget_max:** 20000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** False
- **gender_balance_required:** False

Expected Outcomes

Project results are expected to contribute to all the following expected outcomes:

- Advanced, secure frugal (energy-, resource-, data- and cost-efficient, sustainable) AI tools for real-time grid operations (including protection), market operations (if relevant) and consumer empowerment in distributed energy systems.
- A demonstrated AI development environment integrated with the upgraded digital spine of the energy system and energy-saving consumer applications.
- Scaled-up capabilities for learning at edge nodes and federated systems using agent-based architectures.
- A governance framework and operating model for the Gen(AI)-powered digital spine.

Scope

Building up on previous actions in Horizon Europe (e.g. the digital spine of the energy system in cluster 5 as well as cloud and edge continuum and AI data processing activities in cluster 4) and Digital Europe (Common Reference Framework for energy consumer applications across the EU, Common Energy Data Space), projects are expected to:

1. Validate (demo site and operational grid) and scale secure state-of-the-art specialised frugal (energy- and resource-efficient, data- and cost-efficient, locally deployed and sustainable) open-source (Generative-)AI tools and models for real-time grid (and market, if relevant) monitoring, balancing, forecasting and consumer empowerment in a distributed energy system.
2. Demonstrate an AI development environment within the enhanced digital spine of the energy system, including its digital twin, which should incorporate the functionality of the Common Reference Framework for energy saving applications.
3. Prototype and validate at scale machine learning at the edge node level and federated at swarm level in an agent-based architecture, leveraging new mechanisms for scaling down AI models as well as incremental and reinforcement learning.
4. The project's deliverables should be compatible with the principles of the Common European Energy Data Space and consistent with the EU's Apply AI and

Competitiveness Compass, ensuring that they promote AI adoption and competitiveness in line with EU policies, as well as the EU Digitalisation of the energy sector initiatives.

5. Where relevant and possible, the selected projects are expected to make use of the European AI factories and AI gigafactories.
 6. The selected projects are expected, where available, to make use of and further develop commonly agreed European standards and to actively collaborate and contribute to the work of established and de facto European standards developing organisations and open-source communities and platforms.
- Selected projects are expected to contribute to the BRIDGE209 initiative and actively participate in its activities.

Rank #8 · HORIZON-CL5-2026-10-D6-10

WATCH

Enhanced resilience in multimodal passenger transport through digital technologies and generative and discriminative AI

Fit: 46.46 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 46/100

La call HORIZON-CL5-2026-10-D6-10 ottiene un fit del 46.5% dopo ottimizzazione LP. Il criterio dominante e' impact (contributo 15.7/100). Solver: Optimal, CR AHP: 0.0168.

"Project results are expected to contribute to all the following expected"

"319 This decision is available on the Funding and Tenders Portal, in the reference documents section for"

"Horizon Europe, under 'Simplified costs decisions' or through this link:"

Gap principali da colmare

- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente
- ⚠ TRL aziendale inferiore al requisito della call (target: 6)

Azioni concrete (prossimi 3-6 mesi)

- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)
- ✅ Eseguire pilot dimostrativo per salire di almeno 1 livello TRL (3-6 mesi)

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

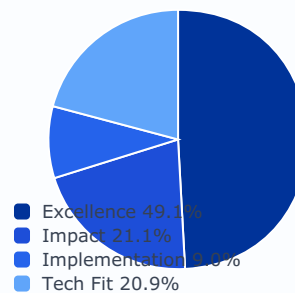
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

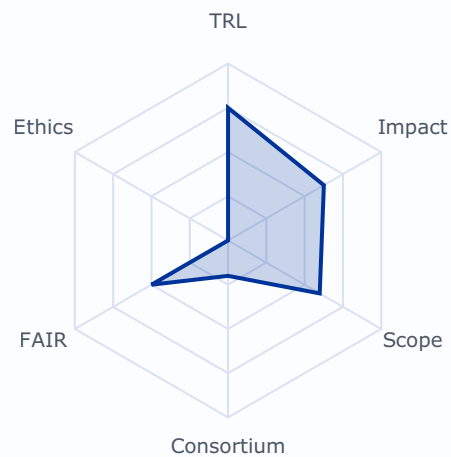
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.6256	0.3000	0.1474
impact	0.2105	0.7439	0.7439	0.1566
implementation	0.0896	0.5300	0.4000	0.0358
tech_fit	0.2086	0.5983	0.5983	0.1248

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** True
- **budget_company_available:** 0.0
- **budget_max:** 10000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** False
- **gender_balance_required:** False

Expected Outcomes

Project results are expected to contribute to all the following expected outcomes:

319 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/lis-decision_he_en.pdf

- Development of systems that utilize real-time data analytics to dynamically respond to disruptions, allowing transport and infrastructure operators to adjust operations swiftly and efficiently;
- Development of predictive maintenance strategies applying innovative Generative AI solutions and utilizing other digital technologies to anticipate and prevent infrastructure and equipment failures;
- Development of guidelines and tools to support passenger transport operators and authorities to ensure minimal disruption in unexpected and critical situations and organise real life emergency simulations;
- Deployment of training programme for transport and infrastructure operators on how to handle transport disruption, including tools and catalogue with contingency planning for specific transport disruptions;
- Reduction of average passenger delay at corridor level (at least 20%) during planned disruptions, compared to the baseline and of (at least 40%) time needed from the generation to the dissemination of a response plan to different stakeholders (e.g. transport operators, passengers, citizens) during unplanned and critical events, compared to the baseline.

Scope

Passenger transport systems are a critical component of urban, sub-urban, and long-distance travel, facilitating connectivity, reducing traffic congestion, and supporting environmental sustainability. However, these systems face challenges such as operational disruptions, safety and security risks, and evolving passenger demands.

In the context of passenger transport, resilience refers to the ability of a transport system to absorb disturbances, maintain its basic structure and function, and recover to a required level of service within acceptable time and cost after being affected by disruption. This involves implementing measures that ensure continuous operation, quick recovery from unexpected events, and adaptability to changing circumstances. Enhanced resilience is essential for maintaining public trust and ensuring smooth functioning of transport systems and it is a cornerstone of the EU Sustainable and Smart Mobility Strategy.

Proposed actions are expected to address all of the following aspects:

- Development and implementation of measures to enhance the resilience of urban, sub-urban and long-distance transport systems.
- Leveraging digital technologies (e.g. big data, digital twins) and in particular generative and discriminative AI for anticipating and predicting the evolution of disruptions and their impacts, with real-time planning and information systems for minimizing impact and enabling faster recovery.
- Use of innovative technologies for data acquisition and integrate data from various sources to inform decision-making and optimise strategies, and generate scenario

libraries for different disruptive events and monitor the implementation of the response plans

- Involve authorities and operators in the design process to create holistic solutions that are user-friendly and aligned with their needs.
- Conducting safety and security assessments in data interpretation, ensuring decisions are based on objective factors while avoiding biases.
- Test and validate the aspects above in real-life use cases in multimodal transport corridor within urban, sub-urban and long-distance passenger transport with minimum three transport modes per use cases (e.g. buses, metros,

Rank #9 · HORIZON-CL5-2026-09-D3-03

WATCH

Innovative technologies and solutions to improve wind energy systems supporting the Strategic Energy Technology (SET) Plan on wind

Fit: 45.19 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 45/100

La call HORIZON-CL5-2026-09-D3-03 ottiene un fit del 45.2% dopo ottimizzazione LP. Il criterio dominante e' impact (contributo 16.4/100). Solver: Optimal, CR AHP: 0.0168.

"Project results are expected to contribute to all the following expected"

"Advance in the achievement of the Strategic Energy Technology (SET) Plan's research"

"Efficient and effective support to wind energy research and innovation through"

Gap principali da colmare

- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente
- ⚠ TRL aziendale inferiore al requisito della call (target: 6)
- ⚠ Requisito SME non soddisfatto dal profilo corrente

Azioni concrete (prossimi 3-6 mesi)

- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)
- ✅ Eseguire pilot dimostrativo per salire di almeno 1 livello TRL (3-6 mesi)
- ✅ Valutare ruolo da partner o struttura legale/cap table compatibile con requisiti SME

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

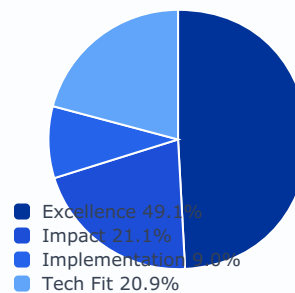
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

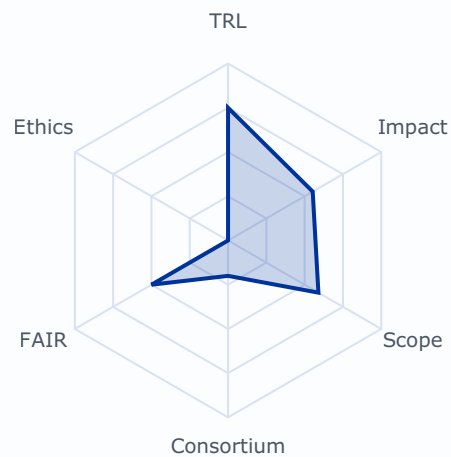
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.5527	0.3000	0.1474
impact	0.2105	0.7770	0.7770	0.1636
implementation	0.0896	0.5300	0.2000	0.0179
tech_fit	0.2086	0.5899	0.5899	0.1231

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** True
- **budget_company_available:** 0.0
- **budget_max:** 93000000.0
- **sme_required:** True
- **sme_ok:** False
- **ssh_required:** False
- **gender_balance_required:** False

Expected Outcomes

Project results are expected to contribute to all the following expected outcomes:

- Advance in the achievement of the Strategic Energy Technology (SET) Plan's research and innovation priorities and targets for wind energy;
- Efficient and effective support to wind energy research and innovation through alignment and cooperation among different actors, avoiding fragmentation of efforts;
- Energy producers and consumers benefit from increased performance of wind energy technologies with the focus on efficiency and flexibility, reduced cost, improved reliability and sustainability, operation and maintenance, safety, robustness and security during all stages of the lifetime of a wind energy farm from installation, operation and maintenance to decommissioning;
- Increased contribution of wind energy to the energy system, with minimal impacts on health and the environment (notably biodiversity and pollution) and at the lowest possible cost.

Scope

The action contributes to address the Strategic Energy Technology (SET) Plan's research and innovation priorities and targets for wind energy, as defined by the Implementation Working Group (IWG) on wind energy, the European Technology and Innovation Platform on wind energy (ETIP Wind)¹⁶⁰ and the European Energy Research Alliance (EERA) joint programme on wind energy¹⁶¹. The activities carried out by the project will pertain to the following priority R&I areas:

- Industrialisation, scale-up and competitiveness: mass production supported by automation and reliable supply chains, design for large volume manufacturing and deployment, design for reliable and lasting products and improved construction and installation methods, digitalisation and artificial intelligence (AI) in industrial environment.
 - Optimisation and further digitalisation of operation and maintenance: integration of artificial intelligence (AI), robotics, advanced repair methodologies and new sensor technologies, interoperability among wind farm sub-systems, advanced communication
- ¹⁶⁰ See for instance: ETIP Wind Strategic Research & Innovation Agenda 2025-2027 - November 2023
- ¹⁶¹ See for instance: NeWindEERA – A New Research Programme for the European Wind Energy Sector – September 2024

technologies and cybersecurity, innovative component replacement solutions and quick connect/disconnect systems for mooring lines and inter-array cables.

- Wind energy system integration: definition and modelling of future system needs, advanced grid capabilities, interoperability, solutions to effectively manage curtailment, integration with energy storage technologies and hybrid plants, distributed wind energy.
- Sustainability and circularity: material substitution for decarbonisation, material reduction, recycling methods, lifetime extension via re-using, refurbishing and re-purposing, new decommissioning tools and methods and solutions to minimise the environmental and biodiversity impacts and optimise the socio-economic impacts.
- Floating offshore wind: development and demonstration of innovative technologies for offshore floating wind. Development of optimised

Rank #10 · HORIZON-CL5-2026-10-D6-06

WATCH

Increasing competitiveness and resilience of multimodal freight transport and logistics for competitive supply chains

Fit: 44.70 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 45/100

La call HORIZON-CL5-2026-10-D6-06 ottiene un fit del 44.7% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 14.7/100). Solver: Optimal, CR AHP: 0.0168.

"Project results are expected to contribute to all the following expected"

"304 This decision is available on the Funding and Tenders Portal, in the reference documents section for"

"Horizon Europe, under 'Simplified costs decisions' or through this link:"

Gap principali da colmare

- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente
- ⚠ TRL aziendale inferiore al requisito della call (target: 7)

Azioni concrete (prossimi 3-6 mesi)

- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)
- ✅ Eseguire pilot dimostrativo per salire di almeno 1 livello TRL (3-6 mesi)

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

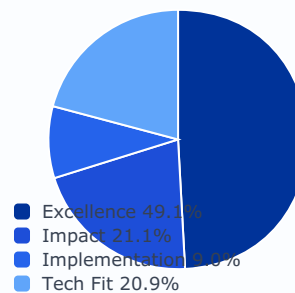
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

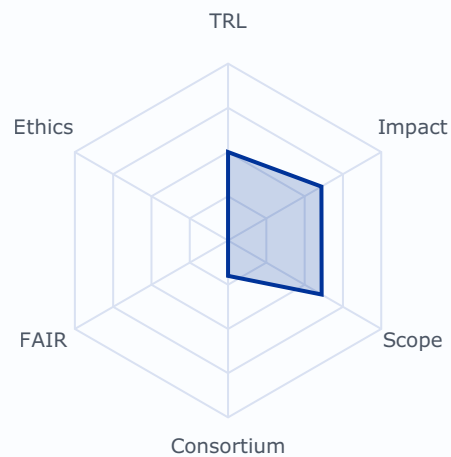
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.6096	0.3000	0.1474
impact	0.2105	0.6561	0.6561	0.1381
implementation	0.0896	0.3800	0.3800	0.0340
tech_fit	0.2086	0.6110	0.6110	0.1275

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** True
- **budget_company_available:** 0.0
- **budget_max:** 20000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** False
- **gender_balance_required:** False

Expected Outcomes

Project results are expected to contribute to all the following expected outcomes:

304 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/lis-decision_he_en.pdf

- Advanced tools and solutions are developed and demonstrated to enhance the competitiveness of European multimodal freight transport networks, ensuring sustainability, resilience, security and increased freight transport capacity;
- Solutions provide real-time visibility, tracking and predictive analytics for multimodal services, performance, and network status (e.g., disruptions, maintenance) to support better planning and increase shippers' adoption of multimodal transport;
- Strengthened integration of multimodal (e.g. rail, inland waterway transport and short sea shipping) data, along with improved synchromodal solutions, enabling more agile, flexible, responsive and resilient multimodal transport logistic systems.

Scope

Freight transport demand in Europe is expected to double in the coming decades, making it essential to enhance the capacity, efficiency, and resilience of freight transport to maintain the European competitiveness. Despite policy goals to strengthen multimodal transport, double rail freight's share by 2050, and increase transport by inland waterways and short sea shipping by 50% by 2050, unimodal road transport remains more competitive, and the growth of multimodal freight has been limited.

The lack of integration of rail, inland waterways and multimodal data into logistics and supply chain solutions is one of the main barriers, hindering the efficient and competitive adoption of multimodality. Furthermore, improving system resilience requires better tools and solutions for multimodal operators to manage disruptions and ensure supply chain continuity.

Building on the results of previously funded research projects and initiatives (including e.g. the Europe's Rail Joint Undertaking), proposals should address all of the following aspects:

- Building on data governance models set in the European Data Act, Common European Data Spaces and the electronic Freight Transport Information Regulation, develop interoperable tools and solutions integrated in existing platforms and solutions (also leveraging the full potential of e.g. Artificial Intelligence, Internet of Things and other emerging technologies) used by shippers, carriers, freight forwarders, warehouse operators, distribution centres, retailers and e-commerce platforms to connect with multimodal information (e.g. services, visibility, ETA, etc.). Developing any type of digital platform or similar is outside the scope of this topic. The proposed tools and solutions should also help operators to react rapidly to disruptions, considering strategies and alternatives for cases of failure, accidents, sabotage, force majeure, etc.
- Map and identify datasets available or to be developed to enhance the integration of multimodal solutions in other supply chain processes (e.g. rail timetable information). Assess how new technologies (e.g. generative Artificial Intelligence, Internet of Things) could be leveraged and applied, including, but not limited to, the procurement and management of smart containers (e.g. for real-time monitoring of cargo conditions,

Rank #11 · HORIZON-CL5-2026-03-D3-20

WATCH

Hybrid AI-Control Framework for a next-generation grid-scale energy storage and system integration

Fit: 42.79 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 43/100

La call HORIZON-CL5-2026-03-D3-20 ottiene un fit del 42.8% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 14.7/100). Solver: Optimal, CR AHP: 0.0168.

"Project results are expected to contribute to all the following expected"

"Advanced digital twin and AI technologies (i.e. hybrid) to support proactive grid storage"

"management and an effective integration with energy and ancillary services markets, for"

Gap principali da colmare

- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente
- ⚠ TRL aziendale inferiore al requisito della call (target: 6)

Azioni concrete (prossimi 3-6 mesi)

- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)
- ✅ Eseguire pilot dimostrativo per salire di almeno 1 livello TRL (3-6 mesi)

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

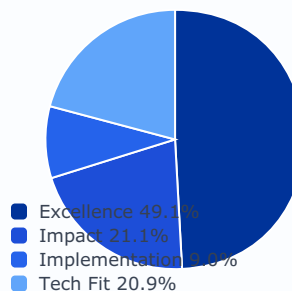
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

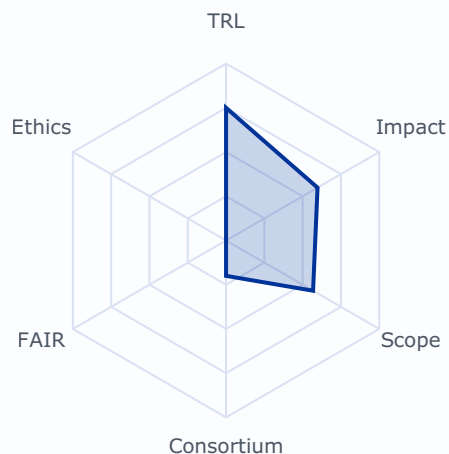
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.5972	0.3000	0.1474
impact	0.2105	0.5993	0.5993	0.1261
implementation	0.0896	0.5300	0.4000	0.0358
tech_fit	0.2086	0.5681	0.5681	0.1185

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** True
- **budget_company_available:** 0.0
- **budget_max:** 14000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** False
- **gender_balance_required:** False

Expected Outcomes

Project results are expected to contribute to all the following expected outcomes:

- Advanced digital twin and AI technologies (i.e. hybrid) to support proactive grid storage management and an effective integration with energy and ancillary services markets, for systems with high levels of renewable energy.
- Optimized energy storage operations resulting in reduced operational and ancillary service costs, extended asset lifespans and increased revenues from market participation while boosting renewable energy uptake for decarbonization.

200 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf

- Optimized energy dispatch and storage cycling, increasing grid resilience and cost efficiency, aiding in lowering energy costs for consumers.

Scope

Projects are expected to develop and validate an integrated system that combines grid-scale energy storage with a hybrid AI control mechanism and optimisation within a digital twin framework. This system will enable proactive and adaptive management of storage and flexibility assets (e.g. large industrial plants with flexible electricity demand), ensure seamless integration with existing grid infrastructures (if applicable also with offshore systems) and electricity markets while enhancing system efficiency and resilience (also for rare/extreme events). The project will demonstrate real-time management for optimal energy dispatch, ancillary services and storage cycling, develop a scalable integration framework for seamless cross-border operability, and validate improvements in grid reliability through field demonstrations and simulations, providing a potential blueprint for EU-wide deployment.

Proof of concept studies using PHIL technology applied to scaled systems are expected to be used for the proactive and adaptive management algorithms.

Selected projects are expected to contribute to the BRIDGE initiative 201 and actively participate in its activities.

Rank #12 · HORIZON-CL5-2027-02-D3-30

WATCH

Advancements in Direct Air Capture

Fit: 42.54 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 43/100

La call HORIZON-CL5-2027-02-D3-30 ottiene un fit del 42.5% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 14.7/100). Solver: Optimal, CR AHP: 0.0168.

"Project results are expected to contribute to the following outcome:"

"Accelerating the commercial availability of Direct Air Capture technologies."

"218 This decision is available on the Funding and Tenders Portal, in the reference documents section for"

Gap principali da colmare

- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente
- ⚠ TRL aziendale inferiore al requisito della call (target: 7)

Azioni concrete (prossimi 3-6 mesi)

- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)
- ✅ Eseguire pilot dimostrativo per salire di almeno 1 livello TRL (3-6 mesi)

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

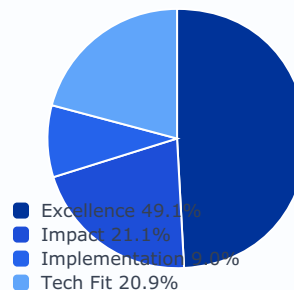
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

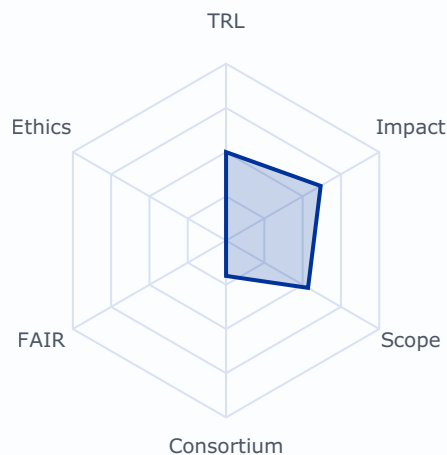
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.6176	0.3000	0.1474
impact	0.2105	0.6284	0.6284	0.1323
implementation	0.0896	0.3800	0.3800	0.0340
tech_fit	0.2086	0.5352	0.5352	0.1116

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** True
- **budget_company_available:** 0.0
- **budget_max:** 16000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** False
- **gender_balance_required:** False

Expected Outcomes

Project results are expected to contribute to the following outcome:

- Accelerating the commercial availability of Direct Air Capture technologies.

217 06700519-1f0f-49b7-bd03-b9021e08fc1b_en

218 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/lis-decision_he_en.pdf

Scope

Projects are expected to aim at scaling up existing technical solutions to at least the level of pilot demonstration in relevant environments (TRL 7 or higher).

Projects are expected to demonstrate:

- Improving the cost efficiency of the capture and regeneration process, including Balance of Plant;
- Improving material performance and process designs (e.g. capture/desorption capacity and kinetics, durability, long-term stability in air);
- Potential for manufacturing and scale-up (recycling and reuse of materials);
- Flexible operation under intermittent or surplus renewable energy supply.

Depending on the technology, other important issues to address are:

- Reducing energy use for sorbent/solvent regeneration;
- Alternative materials for - and approaches to - CO₂ separation;
- Development and demonstration of materials in particular for DAC systems operating in cold temperatures or under humid conditions, or in areas with severe air pollution;
- Accuracy of simulated predictions compared to validation experiments;
- New designs for efficient, high-flux air contactors, optimised for maximum air contact.

Projects must develop a comprehensive life cycle assessment to evaluate the environmental implications of the entire DAC system, from material extraction and use, energy use and greenhouse gas emissions through to end-of-life.

The use of the European Research Infrastructure for CO₂ Capture, Utilisation, Transport and Storage ECCSEL is encouraged but not mandatory. International cooperation with participating countries of the Mission Innovation Carbon

Dioxide Removal Mission²¹⁹ is encouraged.

Rank #13 · HORIZON-CL5-2026-10-D6-09

WATCH

Road Safety and resilience of rural areas

Fit: 41.71 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 42/100

La call HORIZON-CL5-2026-10-D6-09 ottiene un fit del 41.7% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 14.7/100). Solver: Optimal, CR AHP: 0.0168.

"Project results are expected to contribute to all of the following expected"

"316 This decision is available on the Funding and Tenders Portal, in the reference documents section for"

"Horizon Europe, under 'Simplified costs decisions' or through this link:"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance
- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente
- ⚠ TRL aziendale inferiore al requisito della call (target: 6)

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)
- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)
- ✅ Eseguire pilot dimostrativo per salire di almeno 1 livello TRL (3-6 mesi)

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

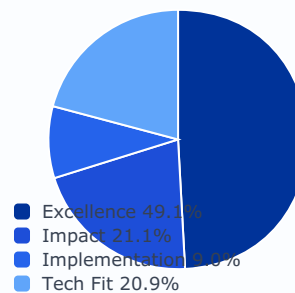
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

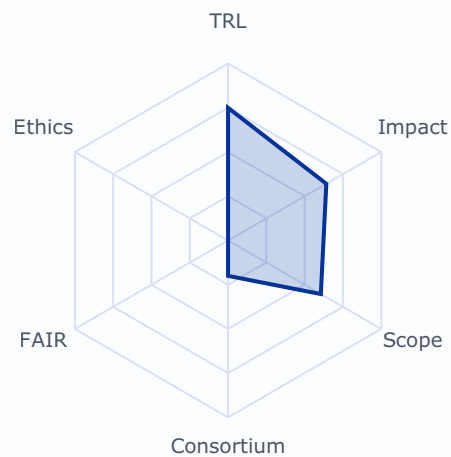
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.6415	0.3000	0.1474
impact	0.2105	0.5105	0.5105	0.1075
implementation	0.0896	0.5300	0.4000	0.0358
tech_fit	0.2086	0.6058	0.6058	0.1264

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** True
- **budget_company_available:** 0.0
- **budget_max:** 12000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** False
- **gender_balance_required:** False

Expected Outcomes

Project results are expected to contribute to all of the following expected outcomes:

316 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/lis-decision_he_en.pdf

- Implementation of the NWRSA methodology for secondary rural roads;
- Innovative and effective enforcement strategies, incentive mechanisms and measures raising risk awareness for fostering safer behaviour;
- Prevention strategies for reducing road fatalities and serious road traffic injuries on rural roads along with the respective implementation guidelines and policy measures tailored to the responsible stakeholders (regional authorities, police, healthcare professionals, national governments, etc.);
- GIS-based application to assist local and regional authorities in identifying and mapping the impact of extreme weather phenomena and other natural disasters (such as floods, fires, storms, heavy snowfall etc.) on the safety and resilience of the road network in their jurisdiction.

Scope

With more than 50% of all EU road fatalities occurring in rural areas along with evidence suggesting that crashes and crash-related fatalities in rural roads vary from those in urban roads or motorways, it is imperative to understand and mitigate the safety risks in rural roads in view of the EU's ambition to move towards Vision Zero by 2050. Local and regional authorities have an important role to play to reduce road fatalities and serious injuries.

In the EU, the Road Infrastructure Safety Management (RISM) Directive introduced the concept of network-wide road safety assessment (NWRSA) and the concept of proactive road safety assessment through the understanding of the in-built safety of roads. Considering safety at the network-level allows for an overview of the road safety performance instead of focusing on isolated parts of it, while the in-built safety assessment aims to identify parts of the road that have been omitted by crash-based analyses (such as crash clusters or hotspot analysis) as they do not concentrate the majority of crashes, yet are crash-prone and/or uncomfortable to navigate. According to the provisions of the RISM Directive, a methodology has been developed to assess the network-wide safety of motorways and primary rural roads based on their combined crash-based and in-built safety assessments. While this is a first step in understanding road safety conditions on rural roads, secondary and lower-class roads are not covered and, at the same time, there is not adequate information on road user behaviour.

In an ageing society, cognitive and physical impairments pose an increasing threat to safe mobility. In rural areas, people with any kind of impairment or disability often lack alternatives to driving a car for their mobility needs. Addressing these issues will not only enhance road safety but also improve the quality of life and prevent the social exclusion of these people.

In addition to road safety issues, local and regional authorities manage risks associated with extreme weather phenomena and other natural disasters like floods, fires, storms or heavy snowfall. As these may affect safety and operations, relevant authorities need to adopt a more holistic resilience monitoring and response.

Research should support addressing these challenges

Rank #14 · HORIZON-CL5-2026-03-D3-22

WATCH

AI-driven forecasting algorithms for Grid and Consumer friendly Energy Sharing – Societal
Readiness pilot

Fit: 41.50 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 42/100

La call HORIZON-CL5-2026-03-D3-22 ottiene un fit del 41.5% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 14.7/100). Solver: Optimal, CR AHP: 0.0168.

"Projects are expected to contribute to all the following expected"

"Develop and test AI-driven forecasting algorithms that use machine learning processes"

"to optimise the value of collective self-consumption through energy sharing for"

Gap principali da colmare

- ⚠ Excellence sotto soglia competitiva: narrativa scientifica da rafforzare
- ⚠ Impact non ancora distintivo su outcomes e policy relevance
- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente
- ⚠ TRL aziendale inferiore al requisito della call (target: 6)
- ⚠ La call richiede copertura SSH esplicita

Azioni concrete (prossimi 3-6 mesi)

- ✅ Definire proof plan tecnico con milestone TRL e metriche di validazione (90 giorni)
- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)
- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)
- ✅ Eseguire pilot dimostrativo per salire di almeno 1 livello TRL (3-6 mesi)
- ✅ Integrare partner SSH e work package dedicato a impatti sociali/adozione

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

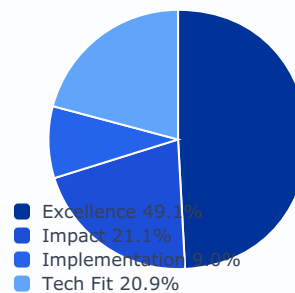
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

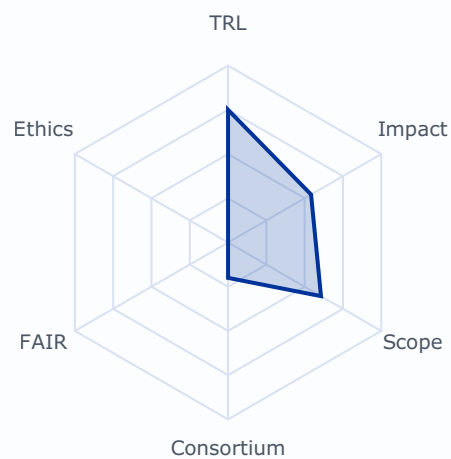
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.5420	0.3000	0.1474
impact	0.2105	0.4998	0.4998	0.1052
implementation	0.0896	0.5300	0.4000	0.0358
tech_fit	0.2086	0.6067	0.6067	0.1266

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** True
- **budget_company_available:** 0.0
- **budget_max:** 12000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** True
- **gender_balance_required:** False

Expected Outcomes

Projects are expected to contribute to all the following expected outcomes:

- Develop and test AI-driven forecasting algorithms that use machine learning processes to optimise the value of collective self-consumption through energy sharing for communities and citizens
- Upscale AI tools and models for grid balancing and forecasting to ensure consumer participation in a distributed energy system.
- Analysis of social interactions using advanced methods like game-theoretic models to highlight the role of active consumers and communities in energy systems.
- Deeper understanding of the needs and concerns of diverse social groups involved in or potentially affected by the R&I development of the technology, thereby increasing the potential for beneficial societal uptake and building trust in results and outcomes.
- Improvement of operation and maintenance of the grid and distributed assets.

Scope

AI-driven forecasting algorithms and machine learning can optimize bidirectional energy transfers involving flexibility services that can help energy communities and jointly acting customers evolve in their role as active participants in the energy system, while ensuring the reduced burdens on DSOs and residual suppliers. Innovations in machine learning forecasting can help balance energy systems and enable active participation in a decentralized, resilient, and digitalized European energy landscape, optimizing self-consumption and reducing energy costs for consumers. This topic is a Societal-Readiness pilot:

- Proposals should follow the instructions applying to the Societal Readiness pilot, as described in the introduction of the Horizon Europe Main Work Programme 2026-2027 for Climate, Energy and Mobility. They entail the use of an interdisciplinary approach to deepening consideration and responsiveness of R&I activities to societal needs and concerns.
- This topic requires effective contribution of the relevant SSH expertise, including the involvement of SSH experts in the consortium, to meaningfully support Societal Readiness. Specifically, SSH expertise is expected to facilitate the socio-technological interface and enable the design of project objectives with Societal Readiness related

Rank #15 · HORIZON-CL5-2026-10-D6-01

WATCH

Flagship-pilot: large-scale demonstrations of CCAM (CCAM Partnership)

Fit: 40.83 / 100 · Solver: Optimal · CR: 0.0168

Perché questa call ha preso 41/100

La call HORIZON-CL5-2026-10-D6-01 ottiene un fit del 40.8% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 14.7/100). Solver: Optimal, CR AHP: 0.0168.

"This flagship pilot is the culmination of the entire activity catalogue"

"carried out by the CCAM Partnership since its launch in 2021. It will combine in one project"

"the most promising CCAM use-cases across three key domains, with the technological"

Gap principali da colmare

- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente
- ⚠ TRL aziendale inferiore al requisito della call (target: 7)
- ⚠ La call richiede copertura SSH esplicita

Azioni concrete (prossimi 3-6 mesi)

- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)
- ✅ Eseguire pilot dimostrativo per salire di almeno 1 livello TRL (3-6 mesi)
- ✅ Integrare partner SSH e work package dedicato a impatti sociali/adozione

Spiegazione Tecnica AHP + LP

Matrice confronto a coppie (AHP)

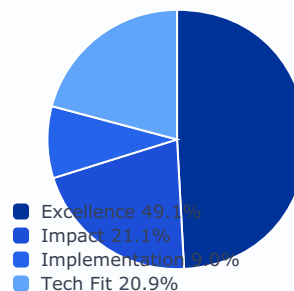
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

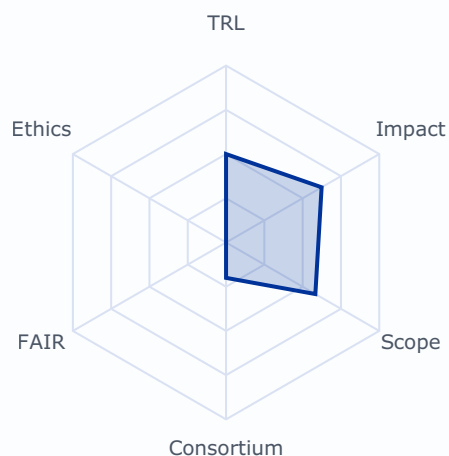
Criterio	Peso AHP	Local pre-LP	Optimal post-LP	Breakdown
excellence	0.4914	0.6239	0.3000	0.1474
impact	0.2105	0.7340	0.5000	0.1052
implementation	0.0896	0.3800	0.3800	0.0340
tech_fit	0.2086	0.5829	0.5829	0.1216

Consistency Ratio: 0.0168 — Matrice consistente

Pesi finali AHP (pie chart)



Spider chart call



Vincoli LP applicati e effetto

- **trl_violation:** True
- **budget_company_available:** 0.0
- **budget_max:** 100000000.0
- **sme_required:** False
- **sme_ok:** False
- **ssh_required:** True
- **gender_balance_required:** False

Expected Outcomes

This flagship pilot is the culmination of the entire activity catalogue carried out by the CCAM Partnership since its launch in 2021. It will combine in one project the most promising CCAM use-cases across three key domains, with the technological advancements from all its R&I clusters²⁷⁴ supporting and enabling CCAM systems and services towards market uptake.

This action is expected to contribute to all the following outcomes:

- Large-scale demonstrations of inclusive, user-oriented, and well-integrated CCAM systems and services for people and goods in mixed traffic through Field Operational Tests (FOTs), Technology Pilots, and Living Labs, building upon advanced and emerging SAE Level 2 systems to move towards SAE Level 3 and 4 functionalities, at multiple test sites and corridors showcasing CCAM potential²⁷⁵, for a minimum of 12 months.
- Validation of enabling technologies that facilitate the extension of Operational Design Domains (ODDs) in large-scale operations and enhance perception performance under poor lighting and adverse weather conditions in large-scale demonstrations and pilots. Assessment of deployment readiness and demonstration of technological maturity focusing on their reliability, security, and real-world applicability.
- Identification of the remaining technological and societal development needs to accelerate deployment and drive user and societal demand. These demonstrations will strengthen the connection with users and society through a co-creative process, ensuring that technological developments align with real-world needs and societal expectations.
- Recommendations for regulatory action aimed at facilitating the deployment of Automated Vehicles (AVs) in Europe, by engaging with relevant policy and regulatory bodies.
- Identification and selection of viable business models for each of the use-cases explored per domain, aiming for continued operation after the flagship pilot through private investment or national/local public funding including mechanisms for transferability and replicability to enable a broader application of results to other cities and regions.

Scope

CCAM solutions are expected to provide a more user-centred, inclusive mobility system that enhances safety, reduces congestion, lowers harmful emissions, and contributes to decarbonization. In addition, CCAM solutions enhance transport effectiveness, thereby strengthening Europe's competitiveness in the global mobility sector. Novel mobility services can enable seamless integration with existing services such as public transport and logistics, while higher levels of automation are expected to boost transport productivity and efficiency. However, the benefits of these solutions must be proven through large-scale demonstrations, validating their effectiveness for both people and goods. It is also of key importance to integrate and test enabling vehicle technologies and to validate trusted communication and

²⁷⁴ <https://www.ccam.eu/our-actions/clusters/>
²⁷⁵ Such as previous experience in testing, developing and deploying CCAM services.

cyber security, as well as real time information transmission. Moreover, a comprehensive assessment of technology maturity is necessary, evaluating the readiness of automated driving functions within mixed traffic conditions and in confined areas. This evaluation helps determine the readiness of automation technologies for deployment, considering factors such as operational reliability, regulatory compliance, and user acceptance. By fostering a systematic approach to large-scale demonstrations, technology validation, and maturity assessment, and by prioritising zero-emission mobility, these efforts contribute to the seamless integration of CCAM solutions across the entire public and private transport ecosystem.

The proposed action is expected to demonstrate different CCAM solutions and technologies in all the following domains:

- Individual mobility within mixed traffic environments, encompassing urban, suburban, motorway, and rural settings, with a focus on the seamless integration of automated and conventional vehicles. Use cases should consider diverse road conditions, infrastructure variability, traffic dynamics, and user needs, ensuring safe, efficient, accessible and user-centric mobility solutions. A key aspect is the integration of advanced technologies (e.g. AI, V2X, cybersecurity, precise positioning, etc.) and



Disclaimer

Report generato automaticamente da Horizon Radar v1.0 – Strumento decisionale basato su Analytic Hierarchy Process (AHP) + Linear Programming (Gurobi). I risultati sono indicativi.

Generative Bionics R&D Team

15/04/2026 01:14